Answer QUESTION 2 PARTS A and B on this page.

PART A

$r(\theta) = 2\sin^2\theta$

$\left.\frac{dr}{d\theta}\right|_{\theta=1.3} = 1.031$

PART B

$\int_{\pi/6}^{-\pi/6} \left((r(\theta))^2 - \left(\frac{1}{2}\right)\right) d\theta$

$\frac{1}{2} = 2\sin^2\theta$

$= 0.2065$

$\theta = -\frac{\pi}{6}, \frac{\pi}{6}$

Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

Answer QUESTION 2 PARTS C and D on this page.

PART C

$\frac{dr}{d\theta} = 4\sin\theta\cos^2\theta - 2\sin^3\theta$

The particle is

PART D

$\frac{dl}{dt} = 15$

$\theta = 1.3$

Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

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